Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Coefficient of linear expansion of brass and steel rods are α_1 and α_2. Lengths of brass and steel rods are ℓ_1 and ℓ_2 respectively. If $(\ell_2 - \ell_1)$ is maintained same at all temperatures, which one of the following relations holds good?

(A) $3\alpha_1\ell_2 = \alpha_2\ell_{1_2}$

(B) $4\alpha_1\ell_2 = \alpha_2\ell_1$

(C) $2\alpha_1\ell_2 = \alpha_2\ell_1$

(D) $\alpha_1\ell_1 = \alpha_2\ell_2$

Q.2

The density of water at $4^\circ\mathrm{C}$ is $1000.0\ \mathrm{kg/m^3}$ and at $100^\circ\mathrm{C}$ it is $958.4\ \mathrm{kg/m^3}$. The cubic expansivity of water between these temperatures is

(A) $4.5 \times 10^{-3}/\mathrm{K}$

(B) $5.4 \times 10^{-5}/\mathrm{K}$

(C) $4.5 \times 10^{-4}/\mathrm{K}$

(D) $5.4 \times 10^{-6}/\mathrm{K}$

Q.3

A steel rod of length $1\ \mathrm{m}$ is heated from $25^\circ\mathrm{C}$ to $75^\circ\mathrm{C}$ keeping its length constant. The longitudinal strain developed in the rod is (Given: Coefficient of linear expansion of steel $= 12 \times 10^{-6}/^\circ\mathrm{C}$)

(A) 6×10^{-6}

(B) -6×10^{-5}

(C) -6×10^{-4}

(D) zero

Q.4

A steel rail of length $5\ \mathrm{m}$ and area of cross-section $40\ \mathrm{cm^2}$ is prevented from expanding along its length while the temperature rises by $10^\circ\mathrm{C}$. If coefficient of linear expansion and Young's modulus of steel are $1.2 \times 10^{-5}\ \mathrm{K^{-1}}$ and $2 \times 10^{11}\ \mathrm{Nm^{-2}}$ respectively, the force developed in the rail is approximately:

(A) $2 \times 10^7\ \mathrm{N}$

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(B) 1×10^5 N

(C) 2×10^9 N

(D) 3×10^{-5} N

Q.5

The density of steel at $0^\circ\mathrm{C}$ is $8.0\ \mathrm{g/cc}$. At what temperature is density lesser by0.1 $\%$? Co-efficient of linear expansion of steel is $10^{-5}/^\circ\mathrm{C}$.

(A) $37.2^\circ\mathrm{C}$

(B) $33.3^\circ\mathrm{C}$

(C) $55.4^\circ\mathrm{C}$

(D) $40.2^\circ\mathrm{C}$

Q.6

A thin steel ring of inner diameter $40\ \mathrm{cm}$ and crosssectional area $1\ \mathrm{mm}^2$, is heated until it easily slides on a rigid cylinder of diameter $40.05\ \mathrm{cm}$. [For steel, $\alpha = 10^{-5}/^\circ\mathrm{C}, \mathrm{Y} = 200\mathrm{GPa}$]
When the ring cools down, the tension in the ring will be :

(A) 1000 N

(B) 500 N

(C) 250 N

(D) 100 N

Q.7

A horizontal tube, open at both ends, contains a column of liquid. The length of this liquid column does not change with temperature. Let γ = coefficient of volume expansion of the liquid and α = coefficient of linear expansion of the material of the tube, then

(A) $\gamma = \alpha$

(B) $\gamma = 2\alpha$

(C) $\gamma = 3\alpha$

(D) $\gamma = 0$

Q.8

When a block of iron floats in mercury at $0^\circ\mathrm{C}$, fraction k_1 of its volume is submerged, while at the temperature $60^\circ\mathrm{C}$, a fraction k_2 is seen to be submerged. If the coefficient of volume expansion of iron is γ_{Fe} and that of mercury is γ_{Hg}, then the ratio k_1/k_2 can be expressed as

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(A) $\frac{1+60\gamma_{Fe}}{1+60\gamma_{hg}}$

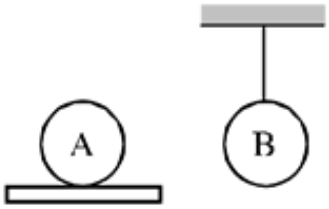
(B) $\frac{1-60\gamma_{\mathrm{Fe}}}{1+60\gamma_{\mathrm{Hg}}}$

(C) $\frac{1+60\gamma_{\mathrm{Fe}}}{1-60\gamma_{\mathrm{Hg}}}$

(D) $\frac{1+60\gamma_{Hg}}{1+60\gamma_{Fe}}$

Q.9

Consider two identical iron spheres, one which lie on a thermally insulating plate, while the other hangs from an insulatory thread. Equal amount of heat is supplied to the two spheres



(A) temperature of A will be greater than B

(B) temperature of B will be greater than A

(C) their temperature will be equal

(D) can't be predicted

Q.10

An iron tyre is to be fitted on to a wooden wheel $1\ \mathrm{m}$ in diameter. The diameter of tyre is $6\ \mathrm{mm}$ smaller than that of wheel. The tyre should be heated so that its temperature increases by a minimum of (the coefficient of cubical expansion of iron is $3.6 \times 10^{-5} /^{\circ}\mathrm{C}$)

(A) $167^{\circ}\mathrm{C}$

(B) $334^{\circ}\mathrm{C}$

(C) $500^{\circ}\mathrm{C}$

(D) $1000^{\circ}\mathrm{C}$

Q.11

If α, β and γ are coefficient of linear, area and volume expansion respectively, then

(A) $\gamma = 3\alpha$

(B) $\alpha = 3\gamma$

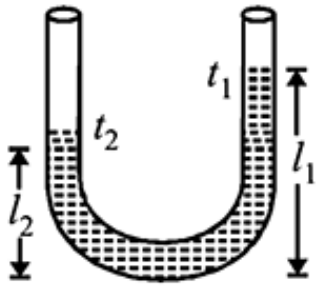
(C) $\beta = 3\alpha$

(D) $\gamma = 3\beta$

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Q.12

In a vertical U-tube containing a liquid, the two arms are maintained at different temperatures t_1 and t_2. The liquid columns in the two arms have heights l_1^1 and l_2^2 respectively. The coefficient of volume expansion of the liquid is equal to



(A) $\frac{l_1-l_2}{l_2t_1-l_1t_2}$

(B) $\frac{l_1-l_2}{l_1t_1-l_2t_2}$

(C) $\frac{l_1+l_2}{l_2t_1+l_1t_2}$

(D) $\frac{l_1+l_2}{l_1t_1+l_2t_2}$

Q.13

Two rods, one of aluminum and the other made of steel, having initial length ℓ_1 and ℓ_2 are connected together to form a single rod of length $\ell_1+\ell_2$. The coefficients of linear expansion for aluminum and steel are α_a and α_s and respectively. If the length of each rod increases by the same amount when their temperature are raised by $t^0\mathrm{C}$, then find the ratio $\ell_1/\left(\ell_1+\ell_2\right)$

(A) α_s/α_a

(B) α_a/α_s

(C) $\alpha_s/\left(\alpha_a+\alpha_s\right)$

(D) $\alpha_a/\left(\alpha_a+\alpha_s\right)$

Q.14

A metallic rod $\ell\mathrm{cm}$ long, A square cm in cross-section is heated through $t^{\circ}\mathrm{C}$. If Young's modulus of elasticity of the metal is E and the mean coefficient of linear expansion is α per degree celsius, then the compressional force required to prevent the rod from expanding along its length is

(A) $\mathrm{EA}\alpha\mathrm{t}$

(B) $\mathrm{EA}\alpha\mathrm{t}/(1+\alpha\mathrm{t})$

(C) $\mathrm{EA}\alpha\mathrm{t}/(1-\alpha\mathrm{t})$

(D) $\mathrm{E}\ell\alpha\mathrm{t}$

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Q.15

A rectangular block is heated from $0^{\circ}\mathrm{C}$ to $100^{\circ}\mathrm{C}$. The percentage increase in its length is 0.10%. What will be the percentage increase in its volume? Answer upto only one decimal place.

(A) 0.03%

(B) 0.1%

(C) 0.3%

(D) 3%

Q.16

Which of the following will expand the most for same rise in temperature?

(A) Aluminium

(B) Glass

(C) Wood

(D) All will expand same

Q.17

A piece of metal weighs $45\ \mathrm{g}$ in air and $25\ \mathrm{g}$ in a liquid of density $1.5 \times 10^{3}\ \mathrm{kg-m^{-3}}$ kept at $30^{\circ}\mathrm{C}$. When the temperature of the liquid is raised to $40^{\circ}\mathrm{C}$, the metal piece weighs $27\ \mathrm{g}$. The density of liquid at $40^{\circ}\mathrm{C}$, is $1.25 \times 10^{3}\ \mathrm{kg-m^{-3}}$. The
coefficient of linear expansion of metal is

(A) $1.3 \times 10^{-3}/^{\circ}\mathrm{C}$

(B) $5.2 \times 10^{-3}/^{\circ}\mathrm{C}$

(C) $2.6 \times 10^{-3}/^{\circ}\mathrm{C}$

(D) $0.26 \times 10^{-3}/^{\circ}\mathrm{C}$

Q.18

A pendulum clock loses $12\ \mathrm{s}$ a day if the temperature is $40^{\circ}\mathrm{C}$ and gains $4\ \mathrm{s}$ a day if the temperature is $20^{\circ}\mathrm{C}$. The temperature at which the clock will show correct time, and the coefficient of linear expansion (α) of the metal of the pendulum shaft are respectively :

(A) $30^{\circ}\mathrm{C}; \alpha = 1.85 \times 10^{-3}/^{\circ}\mathrm{C}$

(B) $55^{\circ}\mathrm{C}; \alpha = 1.85 \times 10^{-2}/^{\circ}\mathrm{C}$

(C) $25^{\circ}\mathrm{C}; \alpha = 1.85 \times 10^{-5}/^{\circ}\mathrm{C}$

(D) $60^{\circ}\mathrm{C}; \alpha = 1.85 \times 10^{-4}/^{\circ}\mathrm{C}$

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Q.19

Three very large plates of the same area are kept parallel and close to each other. They are considered as ideal black surfaces and have very high thermal conductivity. The first and third plates are maintained at temperatures $2T$ and $3T$ respectively. The temperature of the middle plate under steady state condition is

(A) $\left(\frac{65}{2}\right)^{1/4}T$

(B) $\left(\frac{97}{4}\right)^{1/4}T$

(C) $\left(\frac{97}{2}\right)^{1/4}T$

(D) $(97)^{1/4}T$

Q.20

1.56×10^{5} J of heat is conducted through a 12 cm thick wall of area 2 m^2 in one hour. If the temperature difference between the two sides of the wall is 20 °C, then the thermal conductivity (in W m^{-1} K^{-1}) of the material of the wall is

(A) 0.11

(B) 0.13

(C) 0.15

(D) 1.2

Answers & Solutions

Q.1 Answer:

$\alpha_1\ell_1 = \alpha_2\ell_2$

Solution:

From question, $(\ell_2 - \ell_1)$ is maintained same at all temperatures hence change in length for both rods should be same i.e., $\Delta\ell_1 = \Delta\ell_2$

As we know, coefficient of linear expansion, $\alpha = \frac{\Delta\ell}{\ell_0\Delta T}$

$\ell_1\alpha_1\Delta T = \ell_2\alpha_2\Delta T \Rightarrow \ell_1\alpha_1 = \ell_2\alpha_2$

Q.2 Answer:

$4.5\times10^{-4}/\mathrm{K}$

Solution:

$V = V_0(1+\alpha\Delta\theta)$ where $\Delta\theta$ is the difference in temperature, α is the coefficient of volume expansion.

i.e., $\frac{V-V_0}{V_0}\frac{1}{\Delta\theta} = \alpha$

$\alpha = \frac{\frac{m}{d_2}-\frac{m}{d_1}}{m/d_1}\cdot\frac{1}{\Delta\theta} = \frac{(d_1-d_2)}{(d_1d_2)}\times\frac{d_1}{\Delta\theta}$

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$= \frac{(d_1 - d_2)}{d_2} \cdot \frac{1}{\Delta\theta} = \frac{1000 - 958.4}{958.4 \times 96}$
$= 4.5 \times 10^{-4}$ **per degree** K

Q.3 Answer:

-6×10^{-4}

Solution:

Strain developed

$\varepsilon = \alpha \Delta \mathrm{T} = \left(12 \times 10^{-6}\right)(50) = 6 \times 10^{-4}$

Strain will be negative, as the rod is in a compressed state.

Q.4 Answer:

1×10^5 N

Solution:

Young's modulus $= \frac{\text{Thermal stress}}{\text{Strain}} = \frac{\mathrm{F/A}}{\Delta \mathrm{L/L}}$

$Y = \frac{F}{A.\alpha.\Delta\theta} \left(\because \frac{\Delta L}{L} = \alpha \Delta \theta\right)$

Force developed in the rail $\mathrm{F} = \mathrm{YA}\alpha\Delta \mathrm{t} = 2 \times 10^{11} \times 40 \times 10^{-4} \times 1.2 \times 10^{-5} \times 10$

$= 9.6 \times 10^4 = 1 \times 10^5$ N

Q.5 Answer:

$33.3^{\circ}\mathrm{C}$

Solution:

$\rho = \frac{\rho_0}{1+\gamma\Delta\theta} \cong \rho_0(1 - \gamma\Delta\theta); \frac{\Delta\rho}{\rho} = -\gamma\Delta\theta$

$\frac{1}{100} = 3 \times 10^{-5}\Delta\theta; \Delta\theta = \frac{100}{3} = 33.3^{\circ}\mathrm{C}$

Q.6 Answer:

250 N

Solution:

$$\frac{\Delta\ell}{\ell} = \frac{\pi \times 0.05}{\pi \times 40} = \frac{1}{800}$$

$$\mathrm{T} = \mathrm{Y}\frac{\Delta\ell}{\ell} \times \mathrm{A}$$

$$= 200 \times 10^9 \times \frac{1}{800} \times 1 \times 10^{-6} = 250 \text{ N}$$

Q.7 Answer:

$\gamma = 2\alpha$

Solution:

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Let A_0 and At be the areas of cross-section of the tube at temperature 0 and respectively, 1 = length of the liquid column (constant) V_0 and V_t be the volumes of the liquid at temperature 0 and t respectively, $V_0 = \ell A_0 \quad V_t = \ell A_t$

$V_t = V_0(1+\gamma t) A_t = A_0(1+2\alpha t)$

$\therefore V_t = \ell A_0(1+2\alpha t) = V_0(1+\gamma t)$

$= \ell A_0(1+\gamma t)$ or $\gamma = 2\alpha$

Q.8 Answer:

$\frac{1+60\gamma_{Fe}}{1+60\gamma_{hg}}$

Solution:

Correct option is (a) $\frac{1+60\gamma_{Fe}}{1+60\gamma_{Hg}}$

Let V_o and V be the volume of the iron block at $T = 0^\circ C$ and $60^\circ C$ respectively, then we can write

$$V = V_0\,(1+60\gamma_{Fe}) \quad \ldots\ldots\ldots (I)$$

Let V_0' and V' be the volume of the iron block submerged in the mercury at $T = 0^\circ C$ and $60^\circ C$ respectively, then we can write

$$V' = V_0'\,(1+60\gamma_{Hg}) \quad \ldots\ldots\ldots (II)$$

Dividing (I) by (II), we get

$\frac{V}{V'} = \frac{V_o(1+60\gamma_{Fe})}{V_o'(1+60\gamma Hg)}$

$\Rightarrow \frac{1}{k_2} = \frac{1}{k_1}\frac{(1+60\gamma_{Fe})}{(1+60\gamma_{Hg})} \quad \because \frac{V_o'}{V_o} = k_1$ and $\frac{V'}{V} = k_2$

$\Rightarrow \frac{k_1}{k_2} = \frac{(1+60_{YFe})}{(1+60\gamma_{Hg})}$

Q.9 Answer:

temperature of B will be greater than A

Solution:

Temperature of B will be higher because, due to expansion centre of mass B will comedown same heat is supplied but in B, Potential energy is decreased therefore internal energy gain will be more.

Q.10 Answer:

$500^\circ C$

Solution:

Initial diameter of tyre $= (1000-6)\text{mm} = 994$ mm, so initial radius of tyre $R = \frac{994}{2} = 497$ mm

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and change in diameter $\Delta D = 6\ \text{mm}$ so $\Delta R = \frac{6}{2} = 3\ \text{mm}$ After increasing temperature by $\Delta\theta$ tyre will fit onto wheel Increment in the length (circumference) of the iron tyre $\Delta L = L \times \alpha \times \Delta\theta = L \times \frac{\gamma}{3} \times \Delta\theta \quad [\text{As } \alpha = \frac{\gamma}{3}]$

$2\pi\Delta R = 2\pi R\left(\frac{\gamma}{3}\right)\Delta\theta \Rightarrow \Delta\theta$

$= \frac{3}{\gamma}\frac{\Delta R}{R} = \frac{3\times3}{3.6\times10^{-5}\times497} \Rightarrow \Delta\theta \simeq 500^{\circ}\text{C}$

Q.11 Answer:

$\gamma = 3\alpha$

Solution:

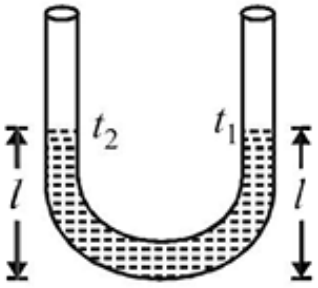
$$V + \Delta V = (L + \Delta L)^3 = (L + \alpha L \Delta T)^3$$
$$= L^3 + \left(1 + 3\alpha\Delta T + 3\alpha^2\Delta T^2 + \alpha^3\Delta T^3\right)$$
$$\Rightarrow \alpha^2 \text{ and } \alpha^3 \text{ terms are neglected.}$$
$$\therefore \quad V(1 + \gamma\Delta T) = V(1 + 3\alpha\Delta T)$$
$$1 + \gamma\Delta T = 1 + 3\alpha\Delta T$$
$$\therefore \gamma = 3\alpha$$

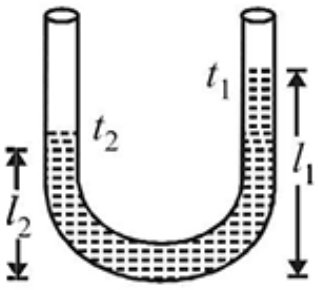
Q.12 Answer:

$\frac{l_1 - l_2}{l_2 t_1 - l_1 t_2}$

Solution:

Suppose, height of liquid in each arm before rising the temperature is l.





With temperature rise height of liquid in each arm increases i.e. $l_1 > l$ and $l_2 > l$ Also $l = \frac{l_1}{1+\gamma t_1} = \frac{l_2}{1+\gamma t_2}$

$\Rightarrow l_1 + \gamma l_1 t_2 = l_2 + \gamma l_2 t_1 \Rightarrow \gamma = \frac{l_1 - l_2}{l_2 t_1 - l_1 t_2}$

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Q.13 Answer:

$\alpha_s/(\alpha_a+\alpha_s)$

Solution:

The lengths of each rod increases by the same amount $\therefore \quad \Delta\ell_a = \Delta\ell_s \Rightarrow \ell_1\alpha_a t = \ell_2\alpha_s t$

$\Rightarrow \frac{\ell_2}{\ell_1} = \frac{\alpha_a}{\alpha_s} \Rightarrow \frac{\ell_2}{\ell_1}+1 = \frac{\alpha_a}{\alpha_s}+1$

$\Rightarrow \frac{\ell_2+\ell_1}{\ell_1} = \frac{\alpha_a+\alpha_s}{\alpha_s} \Rightarrow \frac{\ell_1}{\ell_1+\ell_2} = \frac{\alpha_s}{\alpha_a+\alpha_s}$

Q.14 Answer:

$\mathrm{EA}\alpha\mathrm{t}$

Solution:

$\mathrm{E} = \frac{\mathrm{F/A}}{\Delta//\mathrm{l}} = \frac{\text{stress}}{\text{strain}}$ **where** $\Delta\mathrm{l} = (1'-1) = 1\alpha\mathrm{t}$ **so** $\mathrm{F} = \mathrm{EA}\alpha\mathrm{t}$

Q.15 Answer:

0.3%

Solution:

Given $\Delta\ell/\ell = 0.10\% = 0.001$ **and** $\Delta\mathrm{T} = 100^\circ\mathrm{C}$

Now $\frac{\Delta\ell}{\ell} = \alpha\Delta\mathrm{T}$

or $0.001 = \alpha \times 100$

or $\alpha = 10^{-5}/^\circ\mathrm{C}$

Further $\gamma = 3\alpha = 3\times10^{-5}/^\circ\mathrm{C}$

$\therefore \quad \frac{\Delta\mathrm{V}}{\mathrm{V}} \times 100 = \left(3\times10^{-3}\right)(100) = 0.3\%$

Q.16 Answer:

Aluminium

Solution:

Among glass, wood and metals, metals expand more for same rise in temperature.

Q.17 Answer:

$2.6\times10^{-3}/^\circ\mathrm{C}$

Solution:

$V_1 = \frac{(45-25)g}{d_1}$ **and** $V_2 = \frac{(45-27)g}{d_2}$

$\therefore \quad \frac{V_1}{V_2} = \frac{20}{18} \times \frac{d_2}{d_1}$

or $\frac{V_1}{V_1(1+r_{\text{metal}}\times10)} = \frac{20}{18} \times \left[\frac{d_1}{(1+r_\ell\times10)d_1}\right.$.

After simplifying, we get $\alpha = \frac{\gamma_{\text{metal}}}{3} = 2.6\times10^{-3}/^\circ\mathrm{C}$

Q.18 Answer:

$25^\circ\mathrm{C}; \alpha = 1.85\times10^{-5}/^\circ\mathrm{C}$

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Solution:

Time period of a pendulum,

$$T = 2\pi\sqrt{\frac{l}{g}}$$

where l is length of pendulum and g is acceleration due to gravity. Such as change in time period of a pendulum,

$$\frac{\Delta T}{T} = \frac{1}{2}\frac{\Delta l}{l}$$

When clock gains 12 s, we get

$$\frac{12}{T} = \frac{1}{2}\alpha(40 - \theta)$$

When ciock loses 4 s, we get

$$\frac{4}{T} = \frac{1}{2}\alpha(\theta - 20)$$

Comparing Eqs. (i) and (ii), we get,

$$\begin{aligned} 3 &= \frac{40 - \theta}{\theta - 20} \\ 3\theta - 60 &= 40 - \theta \\ 4\theta &= 100 \\ \theta &= 25^\circ\text{C} \end{aligned}$$

Substituting the value of θ in Eq. (i), we have

$$\begin{aligned} \frac{12}{T} &= \frac{1}{2}\alpha(40 - 25) \\ \frac{12}{24 \times 3600} &= \frac{1}{2}\alpha(15) \\ \alpha &= \frac{24}{24 \times 3600 \times 15} \\ \alpha &= 1.85 \times 10^{-5}/^\circ\text{C} \end{aligned}$$

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Thus, the coefficient of linear expansion in a pendulum clock $= 1.85 \times 10^{-5}/^{\circ}\mathrm{C}$

Q.19 Answer:

$\frac{97}{2}^{1/4}$ T

Solution:

Let temperature of middle plate is given by, T'

Use the condition of steady heat flow,

$\Rightarrow \sigma A 2T^4 + \sigma A 3T^4 = \sigma 2AT'^4$, **where** σ, A **represents Stefan's constant, area of cross-section.**

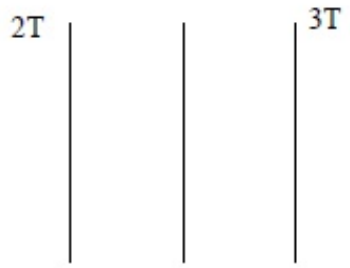
$\Rightarrow 2T^4 + 3T^4 = 2T'^4$

$\Rightarrow 16T^4 + 81T^4 = 2T'^4$

$\Rightarrow 97T^4 = 2T'^4$

$\Rightarrow T'^4 = \frac{97}{2}T^4$

$\therefore T' = \left(\frac{97}{2}\right)^{1/4}T$



Q.20 Answer:

0.13

Solution:

$\frac{dQ}{dt} = \frac{KA\Delta T}{x}$

$\frac{1.56 \times 10^5}{3600} = \frac{K \times 2 \times 20}{12 \times 10^{-2}}$

$\Rightarrow K = \frac{1.56 \times 10^5 \times 12 \times 10^{-2}}{3600 \times 2 \times 10}$ W m^{-1} K^{-1}

$\Rightarrow K = 0.13$ W m^{-1} K^{-1}

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